

Impact Statement -Bridging the Tokenization Gap in African LLMs via SuperBPE

Purpose of the Project

In this project, I am tackling the exclusion of African languages in large language models by designing a tokenizer that reduces bias toward English and other high-resource languages. Current tokenization methods fragment African text inefficiently, which weakens model performance and reinforces inequality. My solution, SuperBPE, allows merges across whitespace boundaries and balances representation across languages. This directly supports my values-led problem statement: building AI that is inclusive, fair, and culturally grounded.

Positive Impacts

The most significant positive impact is linguistic inclusivity. Communities speaking under-represented languages such as Oromo, Tigrinya, and Somali will benefit from language models that tokenize their texts more effectively. This opens opportunities in education, translation, and cultural preservation. Researchers and developers gain reproducible tools to build culturally relevant AI systems, while society benefits from stronger representation of African voices in global technology. Over time, this fosters dignity, autonomy, and community well-being by ensuring African languages are not marginalized in digital spaces.

Potential Risks and Harms

I recognize risks such as reinforcing imbalances if dominant languages continue to overshadow minority ones, or if community contributions inadvertently expose sensitive information. Cultural misrepresentation could occur if texts are tokenized without proper context. Technical risks include overfitting to certain corpora, reducing generalizability. Economically, limited access to computing resources may restrict who can benefit. Environmentally, large-scale training could contribute to energy consumption, though my tokenizer's efficiency helps mitigate this.

Values

This project promotes fairness, inclusivity, transparency, and cultural preservation. Fairness is achieved by balancing corpora across languages. Inclusivity ensures minority languages are represented. Transparency is maintained through open documentation of dataset sources and preprocessing. Cultural preservation is advanced by incorporating community-curated texts such as oral histories and folktales.

Mitigation Strategies

I reduce risks by anonymizing sensitive data, restricting sources to open-licensed corpora, and validating token distributions across languages. Community engagement ensures accountability and helps prevent misrepresentation. Technical risks are mitigated through benchmarking, reproducibility, and open evaluation pipelines. Efficiency reduces environmental impact, while partnerships can expand access to computing resources.

Limitations and Uncertainties

I acknowledge that many African languages remain under-digitized, and synthetic augmentation cannot fully replace authentic community data. There is uncertainty about long-term adoption and sustainability, as broader ecosystem support is required. Despite these limitations, this project provides a foundation for more equitable language technologies.